ME 350R / ME 380R · TEAM BLOCK · WED 14 OCT · ONE SHEET PER TEAM, HAND IN AT THE END

Make your mechanism do the work

You have two actuators. Everything else your robot does, the mechanism has to do. This sheet is how you check that it does.

Team: _____ Track: ☐ hand ☐ wearable ☐ leg

A. Where does your mechanism do the work today? Tick the level your *main* mechanism is at right now. Be honest: this is for you.

	Level	What it means	Examples
☐	**0**	The motor drives the output joint directly. No mechanism.	A servo on a finger joint; a motor on a knee hinge; a wheel instead of a leg.
☐	**1**	A linkage, with dimensions chosen by eye or copied.	"A four-bar that looked right in CAD." Nobody can say why the links are those lengths.
☐	**2**	A linkage **sized by a synthesis method** to a task stated in numbers, then **screened** (assembly, order, Grashof, μ).	A finger dyad through three grasp poses; a tail sized by Freudenstein; a leg sized by today's optimization.
☐	**3**	The mechanism does what two actuators alone could not.	One motor curls three finger joints in a set ratio (coupling); a foot path and timing shaped by the linkage; a grip force multiplied near a toggle; a joint that tracks the knee's moving centre.

At the design freeze (Wed 21 Oct), your main mechanism must be at Level 2 or higher: sized by a method you can name, from a task in numbers, and screened. Level 3 is where the strong projects are. This is not a new rule: it is what "Synthesize" has meant since the first week, made explicit.

B. Move up one level this week. Which linkage will you size, and how?

The task in numbers (poses, angle pairs, or a path with timing): _____

_____

Method: ☐ two rocker limits ☐ three poses ☐ three angle pairs (Freudenstein) ☐ optimization (today)

Who runs it: _____ Dimensions ready by: _____

Rules it must obey, each with a number: $\mu \geq$ _____ motor range _____ pivots must lie in _____

Level 3 question: what could your mechanism do that would take a third actuator otherwise? _____

_____

C. Build it: a mock-up for Wednesday 21 October. Bring a physical mock-up of your main mechanism to the design freeze: cardboard, foamcore, laser-cut or 3D-printed links on pins, at your synthesized dimensions, moving by hand through its task. It will show you, before you build the real thing, what the numbers did not: links that collide, pivots with nowhere to go, a motion you did not expect.

What it will be made of: _____ Who makes it: _____

When and where (the Friday TIW hours, 9–noon, are a good time): _____

The week ahead	
Today, end of day	The design-freeze grading sheet will be posted on Canvas: exactly what you will be graded on.
Mon 19 Oct	In class: running an optimization on the computer, start to finish. Bring a laptop and your task in numbers from Part B.
Wed 21 Oct	**Design freeze.** Main mechanism at Level 2 or higher, with its numbers and its screen; your mock-up.